



Quantum Bluff

USER GUIDE

University of Strasbourg

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1. Introduction & Getting Started

1.1 What is Quantum Bluff?

Quantum Bluff is a multiplayer gaming platform offering:

- **Poker** — cash games and tournaments (Texas Hold'em)
- **Casino games** — Blackjack, Roulette, Slot machines
- **Social systems** — Friends, Loans, Private messages
- **Progression & gamification** — Levels (1–99), Badges, XP
- **Global leaderboards** — by XP, Poker wins, Casino wins

1.2 Platform Access

1.2.1 Available Platforms

Web

Accessible via browser (responsive design, desktop/mobile).

URL: <https://mai-projet-integrateur.u-strasbg.fr/vmProjetIntegrateurgrp10-0/>

Windows Desktop

Local installation from Git repository.

File: **Quantum Bluff Setup 1.0.0.exe** (127 MB)

Location: `client/dist-electron/` from Git repository.

Double-click the executable and follow the installation wizard.

Mac Desktop

Local installation from Git repository.

File: **Quantum Bluff-1.0.2-arm64.dmg** (175 MB)

Location: `Game_Versions/` from Git repository.

Open the DMG file and drag the application to Applications.

Android

Native application via Capacitor.

Source: `client/android/` — Compilation: `npx cap build android`

Transfer the generated APK to your Android device.

iOS

Native application via Capacitor.

Source: `client/ios/` — Compilation: `npx cap build ios`

Build with Xcode and install on your iOS device.

Supported languages: English, French, Spanish, Ukrainian, Arabic
Game currency: Chips (tokens) — *no real money involved.*

1.3 System Requirements

- Stable internet connection
- Modern browser (Chrome, Firefox, Safari, Edge)

- Minimum resolution: 320 px (mobile) to 1920 px+ (desktop)

2. Authentication & Account

2.1 Registration Steps

2.1.1 Getting Started

1. Home screen displays an inspiring quote (random from 20 famous poker quotes).
2. Click **Sign Up** or use **Register**.

2.1.2 Registration Process

Step 1 — Email

- Enter a valid email address.
- System verifies availability.
- Error: `EMAIL_ALREADY_EXISTS` → create a new account or use Login.

Step 2 — Username & Password

- **Username:** 3–20 alphanumeric characters + underscore.
- **Password:** Minimum 8 characters (recommended: uppercase, lowercase, number, special character).
- Password confirmation required.

Step 3 — Security Question

- Select one of 10 secret questions offered.
- Answer with a phrase (case-insensitive).
- Used for account recovery (forgot password).

Step 4 — Date of Birth

- Mandatory for legal validation (minimum age: 18 years).
- ISO format `yyyy-mm-dd` — range: 18 years old to 120 years old.

Step 5 — Activation

- Account created with **1 000 initial chips** (starting balance).
- Automatic redirect to login or lobby.

2.2 Age Validation & Country Restrictions

2.2.1 Country Detection

Detection is based on your request IP + CDN headers (Cloudflare, Vercel, etc.). This is automatic and used only for legal validation during registration.

2.2.2 Age Rules by Country

Country / Region	Min. Age	Notes
United States (US)	21 years	Online gambling federally regulated
All other countries	18 years	International standard
Saudi Arabia (SA)	Forbidden	Online gambling not authorized
Iran (IR)	Forbidden	Online gambling not authorized

2.2.3 Email Blocking System

If you try to register before reaching legal age, your email is **automatically blocked**. You will receive the error: *"This email cannot be used to register before the age eligibility date related to your age."* The email is unlocked **automatically** when you reach legal age (at midnight UTC).

2.2.4 Forbidden Countries

If your IP comes from Saudi Arabia or Iran, registration is **completely refused** (error 403). There is no future unlock system — this is a permanent restriction.

2.2.5 VPN Limitation

If you use a VPN, the detected country is the **VPN exit country** (not the VPN provider location). Example: VPN server in Saudi Arabia but exit in the United States → treated as United States.

2.3 Login

1. **Email:** Enter account email (existence verified).
2. **Password:** Enter password (case-sensitive). Display toggle available (eye icon).
3. Post-login redirect: Poker Lobby by default (or previous page if applicable).

2.4 Account Recovery (Forgot Password)

1. Go to **Forgot Password** screen.
2. Enter email.
3. Answer security question (same answer as at registration).
4. Enter new password + confirmation.
5. Success: redirect to login.

Error codes: `INVALID_EMAIL` (non-existent email), `INVALID_SECURITY_ANSWER` (wrong answer).

2.5 Account Management

2.5.1 Profile (Account Tab)

- **Avatar:** Uploadable photo/avatar (PNG, JPG).
- **Username:** Modifiable (effective after logout/login).
- **Email:** Displayed only (not user-modifiable).
- **Settings:** Language, accessibility, theme.

2.5.2 Logout

Click **Logout** → returns to Auth screen. JWT session is invalidated (token revoked server-side on Redis). Local cache (avatar, gamification, balance) is cleared.

3. Lobby Interface

3.1 Main Navigation Tabs

1. **POKER** (default) — Cash games, Tournaments, Spectate live
2. **MINIGAMES** — Roulette, Slot machines, personal statistics
3. **BLACKJACK** — Solo (1v1 vs dealer) or Multiplayer tables
4. **PROFILE** — Personal statistics, Badges, Levels, Game history
5. **FRIENDS** — Friends list, Pending requests, Messages, Loans
6. **LEADERBOARDS** — Global XP, Poker Wins, Casino Wins

3.2 POKER Section — Available Tables

3.2.1 List Structure

1. **Waiting Rooms** — Display: Name, Small/Big Blind, Players registered, Status. Filters: Public / Private. Buttons: Join / Spectate.
2. **Games In Progress** — Display: Phase (Preflop, Flop, Turn, River, Showdown). Joinable if seat available. Spectate always possible.

3. Tournaments

- *Open* — In registration (shows start date/time, local timezone)
- *In Progress* — Bracket tables live
- *Closed* — Final results

3.2.2 Filters & Search

- **Blind Range:** Slider for min/max big blind
- **Players:** Minimum number of players

- **Status:** Public / Private / All
- **Search:** Table or tournament name

3.3 Creating a Table

Parameter	Details
Name	Table title (visible to others)
Visibility	PUBLIC (all) / PRIVATE (invite only)
Max Players	2–5 (heads-up to 5-max)
Small Blind	e.g. 5, 10, 25
Big Blind	Must be > Small Blind
Min Balance	Minimum chips to play (optional)
Turbo Mode	Fast blind increase (checkbox)

Validation: Minimum minBalance (ou 100 par défaut). Level may restrict available blinds. Max 5 simultaneous tables per player. After creation, the player becomes host and is redirected to the Waiting Room.

3.4 Joining a Table (Cash Game)

1. Select table from lobby list.
2. Enter buy-in (recommendation: 20–100× BB).
3. Upload avatar (optional).
4. Confirm.

Buy-in limits: Minimum minBalance (ou 100 par défaut). Maximum: player balance or table cap. Insufficient balance triggers `INSUFFICIENT_CHIPS` error.

3.5 Creating a Tournament

Parameter	Details
Name	Tournament title
Max Players	4–20 (determines bracket)
Blinds Structure	Micro (5/10), Small (10/20), Medium (25/50), Large (100/200)
Initial Stack	Starting chips per player (e.g. 1500)
Buy-in	Registration cost (chips, deducted from balance)
Start Date/Time	Local format (datetime-local)
Turbo Mode	Blind increase 2× faster

Max 1 tournament per player simultaneously. Status: **Waiting** until start time, then **In Progress**.

3.6 Personal Dashboard Widgets

1. **Balance** — Currently available chips

2. **Level & XP** — Progress bar toward next level
3. **Recent** — Last action (game played, win/loss)
4. **Friends Online** — Avatar + status (Playing / Idle)
5. **Daily Challenge** — Current challenge + progress
6. **Free Top-up** — If promo code **QUANTUM** is valid (+1000 chips, once per day)

4. Poker — Cash Games & Tournaments

4.1 Fundamental Rules (Texas Hold'em)

4.1.1 Variant

Texas Hold'em only. 2–9 players per table. Community cards: 5 (Flop 3 + Turn 1 + River 1). Final hand: best 5 cards from 7 (2 private + 5 community).

4.1.2 Roles & Positions

Heads-up (2 players):

- **Button** = Small Blind (acts last postflop, first preflop)
- **Other player** = Big Blind (acts first postflop)

3+ players:

- **Dealer Button (D)**: Marked by "D" chip, rotates clockwise
- **Small Blind (SB)**: Left of dealer, posts SB before cards
- **Big Blind (BB)**: Left of SB, posts BB before cards
- **Others**: PLAYER (positions UTG, UTG+1, ... CO, Button)

Preflop action order: UTG → UTG+1 → ... → CO → Button → SB → BB

Postflop order: SB (if active) → BB → UTG → ... → Button

4.1.3 Game Phases

1. **Pre-Game** — Players seat, chips distributed
2. **Preflop** — Blinds posted, 2 cards dealt per player
3. **Preflop Action** — Bet/fold/check/raise until equalization
4. **Flop** — 3 community cards revealed
5. **Flop Action** — Bet/check/fold/raise
6. **Turn** — 4th community card
7. **Turn Action** — Same mechanic
8. **River** — 5th and final community card
9. **River Action** — Bet/check/fold/raise

10. Showdown — Best hands compared, pot distributed

4.1.4 Action Options

- **FOLD** — Abandon the hand (chips at stake lost)
- **CHECK** — Pass without betting (only if no one has bet)
- **CALL** — Equalize previous bet
- **RAISE** — Bet more (minimum: big blind or previous raise)
- **ALL-IN** — Commit all remaining chips (can be < required bet)

4.1.5 Hand Rankings (Best to Worst)

#	Hand	Description	Example
1	Royal Flush	A-K-Q-J-10 same suit	A♠K♠Q♠J♠10♠
2	Straight Flush	5 consecutive, same suit	9♥8♥7♥6♥5♥
3	Quads	4 identical cards	K♠K♥K♦K♣
4	Full House	Three of a kind + Pair	J♠J♥J♦6♣6♥
5	Flush	5 cards same suit	K♣9♣7♣5♣2♣
6	Straight	5 consecutive (mixed suits)	8♠7♥6♦5♣4♠
7	Three of a Kind	3 identical + kickers	5♠5♥5♦
8	Two Pair	2 different pairs	Q♠Q♥3♦3♣
9	Pair	2 identical cards	10♠10♥
10	High Card	Highest card (no pair)	A♠K♣Q♦J♥9♠

Tie-breaking: Kickers (unpaired cards) compared if same hand type. Example: Pair of Kings + A-K-Q beats Pair of Kings + A-J-T (better kicker).

4.2 Gameplay — Typical Game Cycle

4.2.1 Before a Hand

1. 2–9 players seated at the table.
2. Blinds automatically debited.
3. 2 private hole cards dealt per player.

4.2.2 Showdown Example

```

Player A: A-K (flush) vs Player B: Q-Q (three of a kind)
Pot: 1200 chips
Winner: Player A (flush > three of a kind)
Balance changes: A +1200, B -1200

```

After each hand: dealer button rotates clockwise, blinds repositioned, cycles repeat.

4.3 Cash Game — Specifics

- **Initial Stack:** Chosen by player at arrival (100–10 000 chips max).
- **Rebuy:** Buy more chips if stack reduced (optional).
- **Leave:** Leave table between hands; chips credited to wallet.
- **Action Timeout:** 30–60 seconds per player; auto-fold if exceeded.
- **Avatars:** Upload avatar on arrival, visible to all players.

4.3.1 Spectator Mode

Access via **Spectate** button from lobby or during game. Spectators see community cards but not other players' hole cards.

4.4 Hidden Bets (Side Bets — Multiplayer)

- Available in multi-player cash games only.
- **Quote** (bet player wins) or **Place** (bet player loses).
- **PRE-HAND window** (5 sec) opens before each hand.
- Between streets (Flop/Turn/River), poker action is frozen.
- At showdown, bets settled and hydrated in balance.
- Recorded in wallet ledger as `HIDDEN_BET_SETTLED`.

4.5 Tournaments

4.5.1 Structure

Before: Registration open → players pay buy-in → initial stack distributed.

During:

- Blind levels increase every 10–30 minutes.
- Players assigned to tables by bracket algorithm.
- Tables reduce as eliminations occur.
- Final table: last 6–9 players.

After: Winner = last player with chips. Payout structure: 1st > 2nd > 3rd...

4.5.2 Payout Structure Example (64 players)

1st:	5000 chips
2nd:	3000 chips
3rd:	2000 chips
4-6:	1200 chips
7-12:	600 chips
13-24:	300 chips
25+:	No payout

4.5.3 XP Rewards

- All participants receive XP (even if eliminated early).
- Bonus: +10 XP per round advanced.
- Top 3: +50 XP | Top 10: +25 XP | 1st Place: +100 XP cumulative.

4.5.4 AFK Management (Between Hands)

After each hand, a 30-second "Ready" window displays for all survivors. Players who do not click **Ready** are auto-eliminated (AFK). If all players are AFK, the player with the most chips wins.

4.6 Bot Tables — Practice & Expert Mode

Difficulty	AI Style	Chips / XP
Easy	Loose / passive	XP earned; chips NOT credited to wallet
Medium	Mixed	XP earned; chips NOT credited to wallet
Hard	Tight / aggressive	XP earned; chips NOT credited to wallet
Expert	Highly trained AI	XP earned; chips credited to REAL wallet

4.7 In-Game Tools

All tools accessible at bottom-right of screen during a game.

4.7.1 CHAT — Emoji & Quick Messages

- **Emoji tab:** 20 predefined poker reactions (e.g. 🤔).
- **Quick messages tab:** 12 preset messages (e.g. "Well played!", "GG WP").
- Visible to all players and spectators.

4.7.2 COMBIS — Hand Rankings Reference

Opens a panel displaying all 11 hand rankings with card examples, color codes, and kicker explanations. Useful for quick reference during play.

4.7.3 PROBAS — Real-Time Probability Display

- Numeric win probability (e.g. "67% win chance").
- Visual gradient bar: green (strong) → yellow (competitive) → red (underdog).
- Current hand display: hole cards + best 5 community cards.
- Updated each street (flop, turn, river).

- Draggable panel; position saved in localStorage.

5. Casino Games

5.1 Blackjack

5.1.1 Objective

Get closer to 21 than the dealer without going over (bust).

5.1.2 Rules

Deck: 6 decks (312-card shoe), automatically shuffled per session. Crypto RNG — no card counting possible.

Card values:

- Numeric 2–10: face value
- Figures (J, Q, K): 10 points
- Ace (A): 11 or 1 (best value ≤ 21)
- *Soft hand*: Hand with Ace counted as 11 (e.g. A+6 = soft 17)

Natural Blackjack: Ace + Figure/10 in 2 cards — beats non-BJ automatically (payout 3:2). If dealer also has BJ $\rightarrow$ Push (bet returned).

Dealer rules: Hit if total < 17 ; Stand if total ≥ 17 (including soft 17).

5.1.3 Payouts

Player BJ vs Dealer non-BJ :	1.5 x bet (plus initial bet)
Player wins normal :	2 x bet
Push (tie) :	1 x bet (refunded)
Player bust :	0 (lose bet)
Dealer bust :	2 x bet

5.1.4 Available Actions

- **Hit** — Draw 1 additional card
- **Stand** — Stop, compare to dealer
- **Double Down** — Double bet, draw exactly 1 card (only with 2 initial cards)
- **Split** — Not available in MVP v1 (future development)

5.1.5 Basic Strategy

Hard Hands (no Ace counted 11):

8-11:	Always double (except vs very strong dealer)
12:	Hit except dealer 4-6 $\rightarrow$ Stand
13-16:	Hit vs dealer 7+ ; Stand vs 2-6

17+: Always Stand

Soft Hands (Ace = 11):

A+2, A+3: Hit
 A+4, A+5: Double if dealer 4-6, else Hit
 A+6: Double if dealer 3-6, else Hit (soft 17)
 A+7: Stand vs 2,7,8 ; Double vs 3-6 ; Hit vs 9-K
 A+8, A+9: Always Stand

5.2 European Roulette

5.2.1 Wheel

37 cases: 0 (green) + 1-36 (red/black alternating).

Red numbers: 1,3,5,7,9,12,14,16,18,19,21,23,25,27,30,32,34,36

Black numbers: 2,4,6,8,10,11,13,15,17,20,22,24,26,28,29,31,33,35

Green: 0

5.2.2 Bet Types & Payouts

Type	Coverage	Payout	Example
Straight	1 number	36:1	Bet "17"
Split	2 adjacent numbers	18:1	"16-17"
Street	3 numbers in a line	12:1	"1-2-3"
Corner	4 numbers square	9:1	"1,2,4,5"
Six Line	6 numbers (2 lines)	6:1	"1-2-3-4-5-6"
Dozen	12 numbers	3:1	"First dozen"
Column	12 numbers (column)	3:1	"Col 1"
Red / Black	18 numbers	2:1	"Red"
Even / Odd	18 numbers	2:1	"Odd"

5.2.3 Limits

Limit	Value
Min bet per line	10 chips
Max bet per line	750 chips
Max total stake	5 000 chips (all bets combined)
Max bets per spin	40 bets

5.2.4 Common Strategies

Martingale (not recommended): Double bet after loss. Risk: losing streak can bust bankroll.

D'Alembert: +1 bet unit after loss, -1 after win. More stable, but house edge still -2.7%.

Outside Bets Only: Red/Black/Even/Odd ($\approx 50\%$ win, 2:1 payout). Lower variance.

5.3 Slot Machine

5.3.1 Symbols, Weights & Payouts

Symbol	Weight	Frequency	Payout (3-of-kind)
Cherry	32	32%	5× bet
Lemon	24	24%	8× bet
Bell	18	18%	10× bet
Seven	14	14%	15× bet
Diamond	8	8%	20× bet

2 identical (pair): 1× bet refunded (break-even).

No match: Bet lost entirely.

5.3.2 Limits

Min bet: 10 chips. Max bet: 1 000 chips (500 at level 1, scaling up to 1 000 at level 25+).

5.3.3 RTP

Theoretical RTP: ~**88–92%**. Over infinite spins, expect to lose 8–12% to the house edge. Each spin is independent (crypto RNG) — no hot streaks.

6. Progression & Rewards System

6.1 Levels & XP

6.1.1 XP Formula

$$XP_threshold(L) = 50 \times L \times (L - 1)$$

Level	Total XP required
1	0
2	100
3	300
4	600
5	1 000
99	482 550 (max)

6.1.2 XP Sources

Source	XP	Condition
Poker Hand (Bot)	12	Preflop played without fold

Source	XP	Condition
Poker Hand Win (Bot)	+20	Win hand vs bots
Poker Showdown Win	35	Cash/tournament showdown win
Poker Showdown Loss	10	Participated (not winning)
Slot Spin	4	Each spin (win or lose)
Slot Win Bonus	+8	Win match (pair+ symbols)
Roulette Spin	4	Each spin
Roulette Win Bonus	+10	Win roulette bet
Blackjack Hand	5	Each hand played
Blackjack Win	+10	Hand won
Login Streak	5/day	Each new daily session

6.2 Badges — Unlock by Level

Level Required	Badge	Description
2	Novice	New player
3	Regular	Regular player
5	Rising	Rising star
7	Skilled	Skilled player
10	Veteran	Veteran
12	Expert	Expert
15	Elite	Elite
18	Master	Master
22	Champion	Champion
25	Legend	Legendary

6.3 Bet Limits by Level

6.3.1 Casino Games

- **Blackjack:** Max 1 000 chips (fixed all levels)
- **Slot Machine:** Level 1 → 500 chips; Level 25+ → 1 000 chips (cap)
- **Roulette:** Max 750 chips/line; Max 5 000 chips total stake

6.3.2 Poker Cash Games — Available Blinds

```

Level 1:    5/10, 10/20
Level 5:    5/10, 10/20, 25/50
Level 10:   5/10, 10/20, 25/50, 50/100
Level 15+:  All blinds available

```

6.4 Daily Challenges & Login Streak

6.4.1 Daily Challenges

Reset at midnight UTC each day. One active challenge per day.

- Examples: "Win 3 poker hands", "Spin roulette 5 times", "Play 2 blackjack hands"
- Reward: +25 XP + 100–200 bonus chips

6.4.2 Login Streak Rewards

Day	XP Reward
1	5 XP
2	10 XP
3	15 XP
4	20 XP
5+	25 XP (cap)

7. Social Interactions

7.1 Friends System

7.1.1 Add a Friend

1. **Friends** tab → **Find Friends** search bar.
2. Type username or email (partial search supported).
3. Click **Add Friend** → recipient receives notification.
4. Recipient: **Pending Requests** → Accept or Decline.

7.1.2 Manage Friends

Actions available on each friend: **Message**, **Invite to Game**, **Block**, **Remove Friend**.
Filter by: Online / Offline / All. Sort by: Recent, Oldest, Alphabetical.

7.2 Private Messages

- Max 500 characters per message.
- URL censoring active (anti-spam).
- History: Last 100 messages per conversation.
- Seen status: Delivered / Seen.
- Notifications: Toast + Friends tab badge on new message.

7.3 Loans Between Friends

7.3.1 Concept

Player A lends chips to Player B. Repayment is automatic: a slice of each win is deducted and sent to the lender until the debt is cleared.

7.3.2 Interest Rates

Repayment Rate	Annual Interest
10%	30%
15%	24%
20%	18%
25%	14%
30%	10%
40%	7%
50%	5%

Total Due formula:

$$TotalDue = Principal + \left(Principal \times \frac{Interest\%}{100} \right)$$

Example: 500 chips @ 20% repayment (18% annual interest) → Total Due = 590 chips.

7.3.3 Auto-Repayment Mechanic

Loan: 500 principal, 20% repayment rate (18% annual)
Total Due: 590 chips

Borrower plays: Bet 50, win 200 → Gross gain = 150 chips
Repayment slice = 150 x 20% = 30 chips
Borrower keeps: 120 chips | Lender receives: +30 chips
Remaining Due: 590 - 30 = 560 chips

7.3.4 Loan Statuses

- **PENDING** — Borrower has not yet accepted
- **ACTIVE** — Loan in progress (interest accumulates live)
- **REPAID** — Fully settled
- **CANCELLED** — Cancelled by lender before acceptance

Limits: Max 1 active loan per borrower. Min amount: 10 chips. Auto-repay stops if borrower balance = 0.

7.4 Invitations

- **From Table:** Menu → Invite Friend → select friend.

- **From Friends Panel:** Click friend → Invite to Game.
- Recipient receives toast notification and auto-navigates to table lobby.

8. Wallet Management

8.1 Balance & Chips

Balance displayed prominently in header. Updated live post-hand. Auto-synced from server every 30 seconds.

8.2 Transaction Types

Type	Description
POKER_CASH_GAME	Poker cash win/loss
POKER_TOURNAMENT	Tournament payout
BLACKJACK_HAND	Blackjack hand result
ROULETTE_SPIN	Roulette spin result
SLOT_SPIN	Slot spin result
HIDDEN_BET_SETTLED	Side bet settled
GIFT_CODE_REDEEMED	Gift code bonus
PROMO_CODE_REDEEMED	Promo code bonus
FRIEND_LOAN_CREATED	Loan principal debit/credit
FRIEND_LOAN_REPAID	Auto-repay debit/credit

8.3 Free Recharge

Promo code: **QUANTUM** → +1 000 chips. Cooldown: 24 hours. Server-side validation (anti-exploit).

8.4 Cheat Prevention

- Server-side balance check before every action.
- Bet validation: min/max caps enforced.
- Crypto RNG: All game results generated server-side.
- Immutable ledger: Transactions recorded irreversibly.
- Rate limiting: API throttling + login max 5 attempts/minute per IP.

9. Tips & Strategies

9.1 Poker — Fundamental Tips

9.1.1 Position Importance

Early Position (UTG, UTG+1): Play tight — top 15% hands (AA, KK, QQ, AK). Many players act after you.

Middle Position (MP1, MP2): Slightly wider — top 20% hands; include mid-pairs (TT, JJ), good aces (AJ+).

Late Position (CO, Button): Widen to top 30% + suited connectors. Act last postflop — biggest advantage.

Blind Positions (SB, BB): Defend with wider range. BB has free look at flop if nobody raises.

9.1.2 Bet Sizing

- Preflop raise: $3\text{--}3.5 \times \text{BB}$ (standard)
- Postflop bet: 50–75% pot
- Raise: minimum $1 \times$ previous bet
- Bluff: less frequent in early stages; increase with deep stacks

9.1.3 Bankroll Management

- Min 20 buyins for cash games; 50 for tournaments.
- Stop loss: walk away after losing 5 buyins.
- Never play above your comfort level.

9.2 Casino — Variance & Expectation

- **Slots:** $\text{RTP} \approx 90\%$ — high variance. Each spin independent.
- **Roulette:** House edge 2.7% on all bets. No bet is statistically "better".
- **Blackjack:** Basic strategy reduces house edge to $\approx 0.5\%$.

9.3 Leveling Efficiently

XP Optimal Farm:

1. Bot Tables: 12 XP/hand (reliable, always available)
2. Roulette Spins: $4 + 10 \text{ XP}$ (if win)
3. Slot Spins: $4 + 8 \text{ XP}$ (if win)
4. Cash Games: 35 XP per showdown win (high variance)

Progression Timeline:

Level 1-5:	Bot tables (quick learning, easy XP)
Level 5-15:	Mix cash games + casino
Level 15+:	Focus cash/tournaments (higher stakes)

Estimated daily XP: Login (+5) + Daily challenge (+25) + 10 bot hands (+120) + Roulette spins (+40–60) = **$\sim 200 \text{ XP/day}$** .

9.4 Milestones & Checkpoints

Level	Milestone	Badge
1–3	First 100 poker hands	Novice → Regular
5	First big casino win	Rising
10	First tournament cash	Veteran
15	1 000 total XP	Elite
25	5 000 total XP	Legend
50	50 000 total XP	Master

10. Interactive Tutorial & Help

10.1 First-Time Experience

Auto-triggered 450 ms after first login. Skip anytime. Duration: ~5–8 minutes (13 lobby steps + 8–10 optional game steps).

10.2 Lobby Tutorial (13 Steps)

Step	Zone	Learning
1	Welcome	Welcome & game objectives
2	Header	Top navigation (profile, settings)
3	Tabs	Poker / Minigames / Blackjack
4	Bot Practice	Training alone vs AI
5	Multiplayer	Join real player tables
6	Waiting Rooms	Understand waiting lists
7	Games in Progress	Spectate live games
8	Minigames	Roulette & Slots quick games
9	Blackjack	Game 21 solo & multiplayer
10	Daily Challenges	Daily challenges for XP
11	Friends	Friends list & private messages
12	Done	Summary & lobby tutorial end

Controls: Next (right chevron) — Previous (left chevron) — Skip (× button).

10.3 Game Tutorial (Optional)

A complete guided heads-up poker hand vs scripted bot (you are BTN+SB, bot is BB). Covers: blinds, hole cards, preflop → flop → turn → river → showdown. Educational spotlight highlights zones with detailed explanations. You can only progress by clicking the expected action.

Access: offered at end of lobby tutorial, or via the ? button at top of lobby.

10.4 Help Access from Anywhere

Look for the **purple circular ? button** at **bottom-left** of screen (fixed position). Opens a help modal with detailed sections. Close with "Got it!" or click outside the modal.

10.5 Tutorial Tips

- No time limit per step — go at your pace.
- Use **Previous** if you missed something.
- After tutorial, all game features are unlocked.
- Tutorial is repeatable anytime from the lobby.

11. Feedback & Game Rating

11.1 Auto-Trigger Rating

After every **5th completed game** (5, 10, 15, ...), a light modal appears at end of game. Completely optional — you can answer "Later".

11.2 Rating Modal

1. **Select rating:** Click 1 to 5 stars (5 = excellent). Stars fill in gold.
2. **Comment (optional):** Up to 2000 characters.
3. **Submit:** Click **Send** to validate, or **Later** to dismiss.

11.3 Manual Rating

Settings button → bottom of modal → Rate the game. Can be done as many times as desired.

11.4 What We Do With Your Ratings

- Ratings sent to `/api/feedback/game-rating` and stored securely.
- Global average tracked over time.
- Rating ≤ 2 stars + comment → support priority (team may contact you).
- Repeated negative feedback → bug analysis initiated.

11.5 FAQ

Q: Are my ratings anonymous?

A: No, they are associated with your account. But your feedback is respected and never used against you.

Q: Can I change my rating after submitting?

A: Currently no. Contact support if you need a follow-up.

Q: Is 5 stars required?

A: You must select at least 1 star to submit. Honest critical ratings (2–3 stars) are equally valuable.

Q: What if I have a bug to report?

A: Ratings < 3 stars with details: include the bug name, when it occurred, and how to reproduce it.

Glossary

All-in

Commit all remaining chips.

Blind

Forced small/big bet posted before cards are dealt.

Bluff

Bet with a weak hand hoping opponent folds.

Buy-in

Chip amount required to play at a table.

Bust Exceed 21 in blackjack.

Call Equalize the current bet.

Chip Virtual game currency unit.

Flop First 3 community cards revealed.

Fold Abandon the current hand.

Heads-up

2-player poker.

Kicker

Unpaired card used to break ties.

Level

Player progression rank (1–99).

Pot Chips accumulated at center of table.

Raise

Increase the current bet.

River

5th and last community card.

Showdown

Cards revealed, winner determined.

Soft Hand

Hand with Ace counted as 11.

Stack

Total chips a player currently holds.

Street

Action round (preflop, flop, turn, river).

Turn 4th community card.

Wallet

Player's personal chip balance.

Support & Troubleshooting

Frequently Asked Questions

Q: Forgot password?

A: Auth menu → "Forgot Password" → answer security question → set new password.

Q: Where to find my avatar?

A: Profile → Edit → Avatar section (upload PNG or JPG).

Q: How to unlock all blinds?

A: Reach level 15+ to unlock all blind levels.

Q: Can I cancel a friend loan?

A: The lender can cancel if the borrower hasn't accepted yet. Once active, auto-repayment continues until settled.

Q: Is the RNG fair?

A: Yes. Crypto RNG is processed server-side (not client-side), verified and seeded.

Contact Support

Email: support@quantumbluff.com

Version & Changelog

Version: 1.0 (MVP) — **Date:** May 2026 — **Last Updated:** 2026-05-16

Implemented Features:

- Poker cash games & tournaments
- Blackjack solo + multiplayer
- Roulette & Slots
- XP / Level / Badges
- Friends & Private Messages
- Friend Loans

- Leaderboards
- Mobile apps (iOS/Android via Capacitor)

Coming Soon:

- 2FA (Two-Factor Authentication) — backend implemented, UI in progress
- Blackjack Split option
- More casino games (Baccarat, Craps)
- Video Poker
- Poker variants (Omaha, Stud)
- Clan/Team system
- Live tournaments
- Sponsorship system

Complete document — Quantum Bluff User Guide v1.0
Share widely & update regularly with new features.